Deliberation or Diversity? A Controlled Study of Multi-Agent AI Feedback for Direct Preference Optimization of Small Language Models

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Abstract—Direct Preference Optimization (DPO) has become the dominant, compute-efficient alternative to reinforcement-learning-based alignment, but it still depends on preference pairs whose collection from human annotators is the single most expensive stage of the alignment pipeline. Reinforcement Learning from AI Feedback (RLAIF) replaces those human labels with model-generated ones, yet it remains unclear which ingredient of a modern multi-agent labelling pipeline actually drives downstream quality. We present the design and reference implementation of a controlled study that isolates two candidate mechanisms: candidate diversity (several personas answering a prompt) and deliberation (a panel of heterogeneous judges debating over two rounds), measured at matched inference compute through a 2×2 factorial design. Every preference set is used to train an identical Qwen2.5-1.5B-Instruct policy with the same LoRA/QLoRA DPO recipe, so the source of the labels is the only variable. We further propose disagreement-aware DPO, which converts inter-judge disagreement into a per-pair label-smoothing term. Evaluation uses a length-controlled win-rate adjudicated by an independent jury of model families that never participated in labelling, alongside rule-checked capability benchmarks, with paired-bootstrap confidence intervals and Holm correction. The entire framework is engineered to run at zero marginal cost on free-tier APIs and a single Google Colab T4 GPU. We describe the architecture, the labelling and evaluation protocols, the reproducibility design, and a reduced-scale pilot that exercises the full pipeline end to end.

Index Terms—preference optimization, RLAIF, multi-agent systems, LLM-as-a-judge, direct preference optimization, parameter-efficient fine-tuning, model alignment

I. Introduction

Reinforcement Learning from Human Feedback (RLHF) [1] and its simpler successor, Direct Preference Optimization (DPO) [2], are the dominant techniques for aligning language models with human preferences. Both consume preference pairs: a prompt, a "chosen" response, and a "rejected" response. Gathering these pairs from human annotators is the costliest part of the alignment pipeline—the reason the HH-RLHF corpus [3] required large-scale paid annotation.

Reinforcement Learning from AI Feedback (RLAIF) [4], [5] replaces human labels with model-generated ones and has been shown to be competitive with human feedback on several tasks. In parallel, a rich toolbox of multi-agent methods—persona ensembles, multi-agent debate [6], and LLM-as-a-judge scoring [7]—has emerged for eliciting higher-quality signals from language models. These components are frequently combined in modern labelling stacks, but their contributions are rarely separated: when a multi-agent pipeline outperforms a single-model baseline, it is seldom clear whether the gain comes from generating a more diverse set of candidate answers or from a more deliberative judging procedure, and whether either survives once inference compute is held constant.

This paper presents the design and open reference implementation of a controlled study built to answer that question for the practically important regime of small language models aligned under a strict zero-cost budget. Our guiding research questions are:

1) Label quality. How accurately do agent-generated preference labels match human labels (HH-RLHF) and verifiable ground truth (GSM8K), and what biases (length, position) do they carry?
2) Mechanism. Is the benefit of a multi-agent pipeline driven by candidate diversity or by deliberation, at matched compute?
3) Downstream effect. How does DPO on agent labels compare with DPO on human labels across model families, sizes, and data scales?
4) Method. Does treating panel disagreement as per-pair label uncertainty (disagreement-aware DPO) narrow the gap to human labels?

Contributions. (i) A 2×2 factorial design that cleanly separates diversity from deliberation in AI feedback at matched inference compute, embedded in a set of reference and adversarial baselines (human, random, longest-answer, self-consistency); (ii) disagreement-aware DPO, a minimal modification that propagates inter-judge disagreement into the DPO loss as a per-pair label-smoothing term; (iii) an evaluation protocol with an enforced judge-independence rule and a length-controlled win-rate that prevents the most common confounds in LLM-as-a-judge

evaluation; and (iv) a fully configuration-driven, resumable, cache-backed implementation that reproduces the entire study at $0 marginal cost on free-tier infrastructure.

II. Background and Related Work

A. From RLHF to DPO

Classic RLHF [1] trains a separate reward model on preference pairs and then optimizes the policy against it with reinforcement learning (typically PPO). This is compute-heavy and unstable to tune. DPO reframes the same objective as a single closed-form, classification-style loss on the policy's log-probabilities of the chosen versus rejected response, removing the reward model and the RL loop [2]. The stability and low cost of DPO are precisely what make preference optimization viable on a single free GPU, where PPO-based RLHF would not be.

B. Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) [8] freezes the base weights and trains small rank-decomposed adapters injected into the projection layers, cutting trainable parameters and memory by an order of magnitude. QLoRA [9] adds 4-bit quantization of the frozen base, so even multi-billion-parameter models fit on a single 16 GB GPU. For the 0.5–3.8B policies studied here, LoRA/QLoRA is what makes DPO feasible on free-tier hardware.

C. AI Feedback and Multi-Agent Methods

Constitutional AI [4] popularized using AI feedback in place of human feedback for harmlessness, and a direct empirical comparison of AI- versus human-generated labels was reported as RLAIF [5]. Multi-agent debate improves factuality and reasoning by having models critique and revise each other's answers [6], while LLM-as-a-judge studies document systematic biases—verbosity, position, and self-preference—that any AI-feedback pipeline must control for [7]. The individual pieces are established; the contribution here is a controlled ablation in which the base model, DPO recipe, and evaluation harness are held fixed and the only variable is the source and construction of the preference labels.

III. Problem Formulation

Let π_θ be the policy initialized from a reference model π_{ref}, and let $\mathcal{D} = \{(x, y_w, y_l)\}$ be a set of preference triples with prompt x, chosen response y_w, and rejected response y_l. DPO minimizes

$$\mathcal{L}_{\text{DPO}} = -\,\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\Big[\log \sigma\big(\beta\,\Delta(x, y_w, y_l)\big)\Big], \qquad (1)$$

$$\Delta(x, y_w, y_l) = \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}, \qquad (2)$$

where σ is the logistic function and β controls the deviation from π_{ref}. Under parameter-efficient training, π_{ref} is the same base network with the LoRA adapter disabled, so no second model need be loaded.

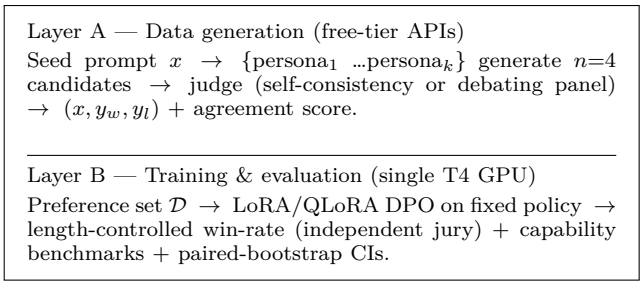


Fig. 1. Two-layer architecture. The boundary between agentic data generation and the ML training core is strict: the only thing that changes between experimental conditions is the preference set produced by Layer A.

TABLE I
Candidate Base Policies (Permissively Licensed Subset)

Model	Params	License
Qwen2.5-1.5B-Instruct (primary)	1.5 B	Apache-2.0
Qwen2.5-0.5B-Instruct	0.5 B	Apache-2.0
SmolLM3-3B	3 B	Apache-2.0
Phi-4-mini-instruct	3.8 B	MIT

The study holds π_θ, π_{ref}, β, and the optimization schedule fixed across all conditions. The experimental variable is the construction of $\mathcal{D}$: whether candidates come from one persona or several, whether the chosen/rejected decision is made by a single self-consistent judge or a deliberating panel, and whether the human-labeled HH-RLHF pairs are substituted as a gold-standard reference.

IV. System Architecture

The system is split into two layers that are kept conceptually and operationally separate (Fig. 1). Layer A (data generation) is agentic and prompt-engineering only—it never touches model weights and performs only LLM API calls. Layer B (training and evaluation) is the machine-learning core: LoRA adapters, the DPO loss, and a held-out evaluation harness. Every parameter of both layers lives in version-controlled YAML configuration files; no experimental constant is hard-coded, and each run writes a snapshot of its fully resolved configuration for provenance.

A. Base Model Selection

The policy must (a) run on a free 16 GB T4 with LoRA, (b) carry a permissive license, (c) follow instructions well enough that DPO refines rather than teaches behavior, and (d) be well supported by the Hugging Face transformers, trl, peft, and Unsloth stack. Table I summarizes the candidates considered. We adopt Qwen2.5-1.5B-Instruct (Apache-2.0) [12] as the primary policy for its balance of capability, license, and training speed, and use Qwen2.5-0.5B-Instruct, SmolLM3-3B, and Phi-4-mini-instruct—all Apache-2.0 or MIT—as a generalization set spanning size and model family.

V. The Multi-Agent Labelling Pipeline

Layer A turns each seed prompt into one preference pair plus a scalar agreement score. It has two factors, corresponding directly to the two hypothesized mechanisms.

A. Factor 1 — Candidate Diversity

For every prompt the generator produces $n = 4$ candidate answers at temperature 0.8. In the single-persona level all four share one neutral helpful-assistant system prompt, so their diversity comes only from sampling noise. In the multi-persona level the four answers come from four distinct personas—concise, step-by-step, domain-expert, and careful-skeptic—each of which nonetheless begins from the same helpful-assistant instruction so that personas vary style and depth, not willingness to help. A shared length instruction ("keep your answer under 200 words") is appended to every persona in both levels, so answer length is never a hidden difference between conditions.

B. Factor 2 — Deliberation

A judge converts the candidate set into a chosen/rejected ordering against a fixed rubric scoring helpfulness, correctness, and harmlessness on a 1–5 scale. The rubric explicitly instructs judges to score substance rather than length or style; length bias is nonetheless measured afterward rather than assumed away. Candidates are shown in a freshly randomized order on every call, and the permutation is logged, so position bias averages out. The two levels of this factor are matched at exactly six model calls per prompt:

- Single judge (self-consistency). One strong judge model scores the candidates in six independent samples; all six are treated as final verdicts.
- Panel (deliberation). Three judges from two model families score independently in round 1, then in round 2 each judge sees the others' round-1 rationales and may revise [6]. Three judges × two rounds = six calls; only the three round-2 verdicts are final.

Each candidate's score is the mean over rubric criteria; the chosen answer is the one with the highest mean over final verdicts and the rejected answer the lowest, with exact ties broken by a seeded coin flip rather than by candidate order. The aggregation rule is identical across both judging levels so it cannot act as a hidden third difference. For every pair, the fraction of final verdicts that agree on the chosen-over-rejected ordering is recorded as an agreement score in $\{1/3, \ldots, 1\}$.

C. Robustness and Cost Engineering

Verdicts are returned as JSON validated with a typed schema and retried on parse failure. Every API request—keyed by model, messages, parameters, and sample index—is hashed and cached in SQLite, so re-running never re-spends free-tier quota and the exact raw outputs behind the dataset are preserved. Requests use exponential backoff with jitter on rate-limit and server errors, and every stage is resumable: a prompt that fails after retries is simply absent from its output file and is picked up on the next run. A dry-run budgeting command reports the per-model API calls and days required for any plan before a single token is spent.

TABLE II
Experimental Conditions

Condition	Candidates	Labels
References and baselines		
base	—	untuned (zero point)
human	HH-RLHF	human (gold standard)
random	multi-persona	coin flip (floor)
length	multi-persona	longer answer
The 2×2 factorial (6 judge calls each)		
single_single	single persona	single judge
single_panel	single persona	panel
multi_single	multi-persona	single judge
multi_panel	multi-persona	panel
Method and variants		
multi_panel_da	multi-persona	panel + disagree.-aware
onpolicy_panel	policy samples	panel

VI. Experimental Design

The core of the study is a set of conditions that share everything except the preference labels (Table II).

The 2×2 interior (single_single, single_panel, multi_single, multi_panel) is the mechanism experiment: diversity on/off crossed with deliberation on/off, all at six judge calls per prompt. single_single is the plain-RLAIF baseline; multi_panel is the full pipeline. The random and length conditions are adversarial baselines: if coin-flip labels train as well as agent labels there is no signal, and if "pick the longer answer" matches the agents, the agent signal is mostly a length preference—something worth knowing before reviewers raise it. An onpolicy_panel variant draws candidates from the policy's own distribution, which often trains better than answers written by a much larger model [10].

The main grid uses 1,000 pairs per condition with three training seeds; a generalization sweep repeats the headline conditions on three further models; and a scaling curve (250/500/1000/2000 pairs, with nested subsets so larger sets strictly contain smaller ones) tests whether the agent–human gap shrinks with scale. All hyperparameters are frozen once by a pilot sweep run jointly on the human and full-agent datasets, so tuning never silently favors one condition.

A. Disagreement-Aware DPO

Panel disagreement is a natural, free estimate of label noise: pairs on which the judges split are more likely mislabeled. We propose replacing the constant DPO label-smoothing term with a per-pair value derived from the recorded agreement. Writing $a_i \in \{1/3, \ldots, 1\}$ for the

agreement on pair i, the conditional DPO (cDPO) objective

$$\mathcal{L}_i = -(1-\epsilon_i)\log\sigma(\beta\Delta_i) - \epsilon_i\log\sigma(-\beta\Delta_i) \qquad (3)$$

uses $\epsilon_i = \epsilon_{\max}(1-a_i)$, so unanimous pairs are trusted fully ($\epsilon_i \to 0$) while contested pairs are down-weighted toward a uniform target. This requires no extra labels or compute beyond the agreement score the panel already produces. Equation (3) defines the method evaluated in the multi_panel_da condition; the current reference implementation flags it explicitly as not yet wired into the trainer, and realizing it is the immediate next step (Section X).

B. Datasets

Seed prompts are drawn from the HH-RLHF helpful-base and harmless-base subsets, stratified 50/50 and restricted to single-turn prompts of 4–256 policy tokens. The same prompts' real human-labeled pairs form the human reference. Exact and near-duplicate prompts (cosine similarity ≥ 0.90 over MiniLM embeddings) are removed so training prompts cannot leak into evaluation. GSM8K supplies a verifiable domain in which a candidate's final answer can be checked against gold, giving an objective label-accuracy measurement independent of any judge.

VII. Evaluation Protocol

Every model—including the untuned base—passes through an identical harness with greedy decoding on the same held-out prompts.

A. Length-Controlled Win-Rate with an Independent Jury

The headline metric is a pairwise win-rate judged by a jury of model families. Each pair is judged twice with the A/B positions swapped; disagreement between the two orders is counted as a tie, removing position bias rather than averaging over it. A strict majority of jurors is required for a win. Crucially, a judge-independence rule is enforced in code: no evaluation juror may share a model family with (a) the policy being evaluated—which would invite self-preference [7]—or (b) any model used to generate or label the training data, since the policy was trained on that judge's taste and would be grading its own homework. Two independent families (Gemini and Cohere) serve as jurors. Because LLM judges favor longer answers, we additionally report a length-controlled win-rate in the style of AlpacaEval 2 [11]: a logistic regression of the verdict on the length difference, read off at zero length difference, reported beside the raw rate so length cannot inflate the headline number.

B. Capability Retention and Statistics

To check that alignment does not degrade core ability, we run IFEval, TruthfulQA-MC2, GSM8K, and MMLU through the EleutherAI lm-evaluation-harness for standard, comparable scoring; the rule- and log-prob-checked tasks need no judge. Uncertainty is quantified with a paired bootstrap over prompts (10,000 resamples): because every model answers the same prompts, resampling prompts rather than individual answers yields honest confidence intervals for differences between models. Results are averaged over three seeds with the seed-to-seed standard deviation reported, and because many conditions are compared, p-values are Holm-corrected to hold the family-wise error rate at 5%.

VIII. Implementation and Reproducibility

The framework is a Python package organized as a thin numbered-script CLI over a library core. Training supports two interchangeable backends resolved automatically from the hardware: Unsloth-patched 4-bit QLoRA on a CUDA T4, and plain PEFT LoRA in full precision on Apple MPS or CPU, so the same experiment code runs on a laptop for development and on a free Colab/Kaggle GPU for the full study. The LoRA configuration adapts all attention and MLP projections at rank 16 ($\alpha = 32$), trained for two epochs at learning rate 5×10^{-6} with an effective batch size of 16 and $\beta = 0.1$; adapters (tens of MB) are checkpointed every 25 steps so an interrupted free session loses at most minutes of work. Reproducibility is enforced structurally: all randomness is seeded, every run snapshots its resolved configuration, preference sets are built over the intersection of prompts valid in all conditions (and locked by a manifest so changing the condition set cannot silently alter which prompts a model trains on), and a test suite covers parsing, schema validation, and the statistics without needing network access or API keys. The upper-bound compute for the full study is roughly 90 GPU-runs (≈60–80 T4 GPU-hours), all within free-tier limits.

IX. Auxiliary Reward Model and Explainability

Before any policy is trained, we ask a cheaper diagnostic question: which observable properties of a response predict that it is preferred? If a label source can be predicted largely from superficial cues such as length, a policy trained on it is likely to learn those cues. We therefore fit an interpretable pairwise reward model to each preference set.

A. Model

Each response y is mapped to a feature vector $\phi(y)$ comprising five named text signals—log word count, hedge-phrase count, refusal-phrase count, question marks, and list items (matched on whole words)—plus the first eight principal components of a MiniLM sentence embedding. Following the Bradley–Terry model [13], $P(y_c \succ y_r) = \sigma\big(w^\top[\phi(y_c) - \phi(y_r)]\big)$ is fitted by L2-regularized logistic regression on standardized feature differences, with no intercept so that the score is antisymmetric in its two arguments. Pairwise accuracy is estimated out-of-fold with 5-fold cross-validation (PCA refitted within each training fold) and reported with a bootstrap 95% CI,

TABLE III
Pairwise reward model on HH-RLHF human labels (5-fold CV). Len. base: accuracy of "the longer response wins".

Pairs	n	Acc. [95% CI]	Len. base
All, text only	2000	52.7% [50.5, 54.8]	51.8%
All, text + PCs	2000	56.5% [54.3, 58.7]	51.8%
helpful-base	1000	67.2% [64.3, 70.1]	63.0%
harmless-base	1000	62.9% [60.0, 65.8]	40.6%

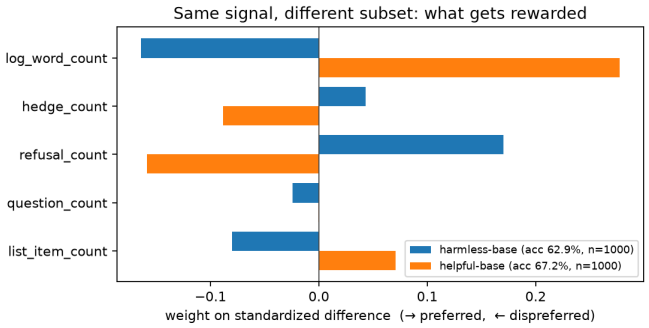


Fig. 2. Standardized reward-model weights on the five named text features, fitted separately per HH-RLHF subset. Length and refusals flip sign between subsets.

beside a "longer response wins" baseline. Attributions are computed with linear SHAP [14]; for each embedding component, the responses at its two extremes are inspected so the axis can be named.

B. Results on Human Labels

Table III reports the model on 2,000 HH-RLHF training pairs. Pooled over both subsets, the model barely clears the length baseline. Fitted per subset, however, accuracy rises to 63–67%, and Fig. 2 shows why: the two subsets reward opposite behavior. In helpful-base, longer answers are preferred and refusals penalized; in harmless-base, shorter answers and refusals are preferred, and the length baseline falls well below chance (40.6%). In the pooled model, the strongest single direction (embedding PC8, coefficient −0.21) separates eager compliance (e.g. "Sure, here's a list of profanities") from cautious, deferring answers, and is penalized; an empathy direction (PC6, "I'm sorry to hear that…") is rewarded. Two implications follow for the main study: AI judges must be checked separately on helpfulness and harmlessness prompts, since a judge with a single global length preference would agree with humans on one subset and disagree on the other; and the length-controlled win-rate of Section VII is warranted, since length is a first-order preference signal in this data.

X. Pilot and Current Status

A reduced-scale "demo" profile exercises the entire pipeline end to end—the same code, prompts, judges, and DPO loss as the full study—using a 0.5B policy, 150 pairs, and the base, human, and multi_panel conditions, so that a complete run (label → train → evaluate) finishes on a laptop in about an hour. This pilot validates the orchestration, caching, resumability, and evaluation plumbing; its numbers are a smoke test, not the study's results, and we therefore do not report them as findings.

Label-agreement pilot. As a first label-quality check, the panel judged 50 held-out HH-RLHF pairs from the label_agreement split, shown in shuffled order, and we compared its choice with the human label (a panel tie counts as half). The panel agreed with the human label on 68.0% of pairs (bootstrap 95% CI 56–80%; 69.6% on harmless-base, 66.7% on helpful-base). Agreement was strongly tied to panel consensus: 80.6% when the final verdicts were unanimous (n=31) against 46.2%, no better than chance, when they were split (n=13); six pairs were scored as ties. This is the pattern disagreement-aware DPO is designed to exploit: inter-judge disagreement flags the labels least likely to be right. Judges named the first-shown response best in 48.7% of 300 calls, so the shuffling leaves no detectable position bias. Deliberation changed individual judges' agreement only modestly (round 1 → round 2: 70 → 74%, 70 → 68%, 64 → 66%). With n=50 these are pilot estimates; the full check uses all 500 pairs. A qualitative inspection of pilot agent pairs also shows the limit of consensus: for a factual lookup prompt, all four candidates gave fabricated addresses and the panel unanimously rated one of them 5/5, which motivates the GSM8K check against gold answers.

Two items remain before the full grid: wiring the disagreement-aware loss of Eq. (3) into the trainer, and executing the main grid, generalization sweep, and scaling curve on free GPU time. This paper documents the design and reference implementation so that the subsequent empirical results are interpretable and reproducible.

XI. Threats to Validity

Construct validity. LLM-judge win-rate is a proxy for human preference; we mitigate its known failure modes through the judge-independence rule, position swapping, and length control, but it cannot fully substitute for a human study. Internal validity. Matching the two factors at six calls per prompt controls for raw compute, yet "one strong judge sampled six times" and "three judges over two rounds" still differ in model identity; the single-judge and panel pools are chosen from the same free catalogue to keep this gap small. External validity. Findings are scoped to small (0.5–3.8B) instruction-tuned policies and to the free model catalogue available at the time of the runs, which changes frequently; the generalization sweep across families and sizes is intended to probe how far conclusions travel. Data validity. HH-RLHF prompts and the free judge models carry their own biases; GSM8K provides an independent, verifiable cross-check on label accuracy.

XII. Conclusion

We have presented the design and open reference implementation of a controlled study that asks not merely whether multi-agent AI feedback helps DPO on small language models, but why—separating candidate diversity from deliberation at matched compute, anchoring both against human, random, and length baselines, and proposing a disagreement-aware DPO loss that recycles panel disagreement as free label-uncertainty information. The framework is engineered end to end for zero-cost reproducibility on free-tier infrastructure, with an evaluation protocol hardened against the standard confounds of LLM-as-a-judge measurement. Completing the empirical grid—and determining which of diversity or deliberation, if either, is the operative mechanism—is the direct next step.

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